| Exp No: 15 | Analyse various types of servers using Webalizer tool |
|---|---|

## AIM:

To analyse the log files of various types of servers (web, proxy and FTP servers) using the Webalizer web server log analysis tool and to study the usage statistics generated by it.

## ALGORITHM / PROCEDURE:

1. Install Webalizer in Linux using `sudo apt install webalizer apache2`.
2. Locate the log files of the servers to be analysed (Apache, Nginx, IIS, FTP, Squid) as given in the table below.
3. Edit the configuration file `/etc/webalizer/webalizer.conf` and set `LogFile`, `LogType`, `OutputDir` and `HostName`.
4. Run Webalizer for each server. The option `-n` gives the host name for the report, `-o` gives the output folder and `-F` gives the log type (clf, w3c, ftp, squid).
5. Open the generated `usage_YYYYMM.html` file in a web browser (`http://localhost/webalizer/apache/`) to see the report.
6. Study the monthly summary, the daily graph, the top URLs, the top clients, the user agents and the HTTP response codes.
7. Compare the results of the different servers and note down the inference.

### Servers and their log formats

| Server type | Log file | Log format | Webalizer option |
|---|---|---|---|
| Apache web server | `/var/log/apache2/access.log` | Common Log Format (CLF) | `-F clf` |
| Nginx web server | `/var/log/nginx/access.log` | CLF compatible (combined) | `-F clf` |
| IIS web server | `C:\inetpub\logs\LogFiles\W3SVC1\u_ex261008.log` | W3C extended | `-F w3c` |
| FTP server (vsftpd) | `/var/log/xferlog` | wu-ftpd xferlog | `-F ftp` |
| Squid proxy server | `/var/log/squid/access.log` | Squid native | `-F squid` |

**webalizer.conf (important lines)**

```
LogFile        /var/log/apache2/access.log
LogType        clf
OutputDir      /var/www/html/webalizer/apache
HostName       apache-server
PageType       htm*
PageType       php
PageType       cgi
HideSite       localhost
```

**Sample log records**

```
Apache (CLF):
192.168.1.25 - - [08/Oct/2026:10:41:07 +0530] "GET /index.html HTTP/1.1" 200 4521
192.168.1.12 - - [08/Oct/2026:10:41:09 +0530] "GET /css/style.css HTTP/1.1" 304 -
10.0.2.2 - - [08/Oct/2026:10:42:15 +0530] "GET /missing.html HTTP/1.1" 404 492
IIS (W3C):
#Fields: date time s-ip cs-method cs-uri-stem sc-status sc-bytes
2026-10-08 05:12:44 192.168.1.30 GET /default.aspx 200 6120
```

## DIAGRAM:

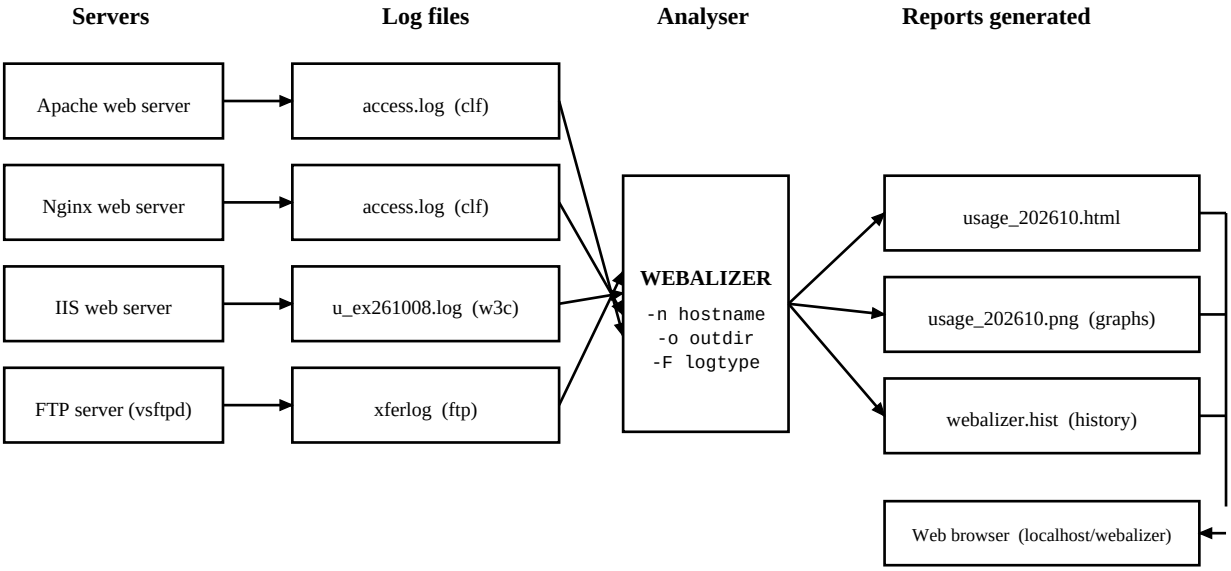


**Fig 1: Log analysis of different servers using Webalizer**

## OUTPUT:

```
USER @USER -VirtualBox: ~
USER @USER -VirtualBox:~$ sudo webalizer -n apache-server -o /var/www/html/webalizer/apache -F clf /va
  r/log/apache2/access.log
Webalizer V2.23-08 (Linux 5.15.0-91-generic) English
Using logfile /var/log/apache2/access.log (clf)
Creating output in /var/www/html/webalizer/apache
Hostname for reports is 'apache-server'
History file not found...
Generating report for October 2026
Generating summary report
Saving history information...
1243 records in 0 seconds
USER @USER -VirtualBox:~$ sudo webalizer -n nginx-server -o /var/www/html/webalizer/nginx -F clf /var/
  log/nginx/access.log
Using logfile /var/log/nginx/access.log (clf)
Creating output in /var/www/html/webalizer/nginx
Generating report for October 2026
Generating summary report
Saving history information...
856 records in 0 seconds
USER @USER -VirtualBox:~$ sudo webalizer -n iis-server -o /var/www/html/webalizer/iis -F w3c u_ex26100
  8.log
Using logfile u_ex261008.log (w3c)
Creating output in /var/www/html/webalizer/iis
Generating report for October 2026
Generating summary report
Saving history information...
522 records in 0 seconds
USER @USER -VirtualBox:~$ sudo webalizer -n ftp-server -o /var/www/html/webalizer/ftp -F ftp /var/log/
  xferlog
Using logfile /var/log/xferlog (ftp)
Generating report for October 2026
64 records in 0 seconds
```

### Report of the Apache web server

| Monthly Statistics for October 2026 | |
|---|---|
| Total Hits | 1243 |
| Total Files | 1131 |
| Total Pages | 402 |
| Total Visits | 138 |
| Total KBytes | 18642 |
| Total Unique Sites | 57 |
| Avg / Max Hits per Day | 155 / 276 |
| Avg / Max Files per Day | 141 / 252 |
| Avg / Max Pages per Day | 50 / 93 |
| Avg / Max Visits per Day | 17 / 23 |
| Avg / Max KBytes per Day | 2330 / 3342 |

## Top URLs by hits

| # | Hits | % | URL |
|---|---|---|---|
| 1 | 214 | 17.2% | /index.html |
| 2 | 188 | 15.1% | /css/style.css |
| 3 | 163 | 13.1% | /images/logo.png |
| 4 | 97 | 7.8% | /login.php |
| 5 | 76 | 6.1% | /about.html |
| 6 | 52 | 4.2% | /contact.html |
| 7 | 41 | 3.3% | /favicon.ico |

## Top clients

| # | Hits | % | Client (site) |
|---|---|---|---|
| 1 | 311 | 25.0% | 192.168.1.25 |
| 2 | 204 | 16.4% | 192.168.1.12 |
| 3 | 142 | 11.4% | 10.0.2.2 |
| 4 | 96 | 7.7% | 192.168.1.30 |

## Hits by response code

| HTTP response code | Hits | % |
|---|---|---|
| 200 - OK | 1019 | 82.0% |
| 304 - Not Modified | 112 | 9.0% |
| 404 - Not Found | 74 | 6.0% |
| 301 - Moved Permanently | 21 | 1.7% |
| 403 - Forbidden | 9 | 0.7% |
| 500 - Internal Server Error | 8 | 0.6% |

### Comparison of the servers

| Server | Records | Total hits | Visits | KBytes | Most requested URL |
|---|---|---|---|---|---|
| Apache | 1243 | 1243 | 138 | 18642 | /index.html |
| Nginx | 856 | 856 | 97 | 12408 | /api/status |
| IIS | 522 | 522 | 61 | 7915 | /default.aspx |
| FTP (vsftpd) | 64 | 64 | 9 | 41230 | /pub/notes.pdf |

### Inference

- The Apache server is the busiest server and its highest load was on 08-Oct-2026 (276 hits).
- About 82% of the requests were successful (200) and 6% gave 404 errors, which means some links on the site are broken.
- The home page `/index.html` is the most visited page. Image and CSS files form most of the hits, whereas only 402 hits are actual pages.
- The same tool can analyse all the servers because only the log type option (`-F`) changes.

## RESULT:

Thus the log files of various types of servers (Apache, Nginx, IIS, FTP) were analysed using the Webalizer tool and the usage statistics such as hits, visits, top URLs, clients and response codes were studied.